

Irfan Ahmad, PhD

Intelligent Science and Technology Program, Hainan University, Haikou, China

☎ +86 186 5075 1364 ✉ Irfan.Ahmad@waikato.ac.nz 🌐 drirfanahmad.com 📄 [ResearchGate](https://orcid.org/0000-0001-9411-4111)

Research Interests

Pattern Mining / Machine Learning / Deep Learning & Computer Vision / Nonlinear Dynamical Systems & Chaos Theory / AI in Education

Profile Summary: Dr. Irfan Ahmad is a computer science educator and researcher with over eight years of international academic experience across New Zealand, Ireland, China, and Thailand. He has taught a wide range of computing and engineering courses, contributed to curriculum development and accreditation activities, and published research in machine learning, computer vision, and nonlinear dynamical systems. He enjoys helping students build strong problem-solving skills, prepare for postgraduate studies and industry careers, and adapt to the opportunities and challenges of modern AI technologies. He currently serves as a Professional Teaching Fellow at the University of Waikato, delivering a joint Computer Science programme at Hainan University, China.

Education

Doctor of Philosophy in Engineering and Technology

Jan 2018 – Dec 2020

Sirindhorn International Institute of Technology (SIIT), Thammasat University

Thailand

- **Department:** Electrical and Computer Engineering
- **Thesis:** *Simple chaotic jerk, hyperjerk, and hyperchaotic hyperjerk circuits and systems: Families of self-excited and hidden attractors.*
- **Supervisor:** Prof. Dr. Banlue Srisuchinwong **Funding:** Fully funded doctoral scholarship
- **Research Areas:** Nonlinear Dynamics — Chaos-Based Secure Communications — Time Series Analysis

Master of Science in Engineering and Technology

Jan 2016 – Dec 2017

Sirindhorn International Institute of Technology (SIIT), Thammasat University

Thailand

- **Department:** Electrical and Computer Engineering **CGPA:** 3.92/4.00 (highest in cohort of 40+ students)
- **Thesis:** *A simplest circuit for the highest complexity of 4D minimum-seven-term no-equilibrium hyperchaos with global (hidden) attractor, and a simple 4D two-transistor chaotic circuit.*
- **Supervisor:** Prof. Dr. Banlue Srisuchinwong **Funding:** Fully funded master's scholarship

Bachelor of Science in Electrical Engineering

Sep 2011 – Aug 2015

Government College University Faisalabad

Pakistan

- **Specialization:** Telecommunication Engineering **CGPA:** 3.97/4.00
- **Distinction:** Gold Medalist — highest-ranking graduate among 100+ students in the degree cohort

Academic Appointments

Professional Teaching Fellow in Computer Science

Mar 2025 – Present

University of Waikato / Hainan University Joint Program

Haikou, China

- Deliver CS courses to 100+ Chinese students per cohort entirely in English; maintain 95%+ pass rate across all courses through structured, skill-focused teaching that connects theory with practical implementation.
- Earn 100% positive teaching evaluations in all courses; students specifically cite step-by-step explanations and lecture quality as key factors in connecting theory with practical implementation.
- Design open-ended labs that require students to diagnose failures, validate outputs, and refine models through critical thinking; one Computer Vision cohort independently collected 2,000+ hand-gesture images and trained transfer-learning models on self-curated data.
- Increase classroom participation by 60–70% through in-class competitions and group presentations despite language barriers.
- Develop and deploy a custom autograding pipeline parsing `.ipynb`, `.html`, and `.py` submissions against a structured rubric, generating per-student feedback at scale.
- Currently develop *CodeVouch*, an AI integrity tool that distinguishes genuine code comprehension from unreviewed AI-generated submissions — addressing an emerging gap in CS assessment.
- Organize professional development seminars on scholarships and career readiness (50+ attendees) — guiding Chinese students on global application processes, documentation, and preparation.

Lecturer / Assistant Professor

Oct 2023 – Feb 2025

Maynooth University (Ireland) at MIEC, Fuzhou University

Fuzhou, China

- Delivered Algorithms and Data Structures, Digital Circuits, and Project-Based Engineering modules to 300+ students, maintaining 90%+ pass rates.
- Increased participation in lectures of 150+ students through real-time polling, instant feedback, and interactive classroom activities.
- Designed and assessed coursework, midterms, and finals under Maynooth University quality assurance processes, including external review and grade moderation.
- Mentored multiple student project teams and supervised a final-year ML project forecasting air quality in Fuzhou using historical environmental data through 2030.

Postdoctoral Associate — Part-time

Aug 2022 – Jul 2023

School of ICT, Sirindhorn International Institute of Technology (SIIT), Thammasat University

Thailand

- Discovered the first coexisting hidden attractors in the smooth cubic Chua's circuit, revealing new parameter regions and multistability phenomena in the system.
- Applied computational search techniques and numerical simulations using MATLAB, Python, and PowerBASIC to analyze attractor dynamics, basin structures, and equilibrium transitions.
- Published findings in a Q1-ranked journal: *International Journal of Bifurcation and Chaos* (IF 2.3, 2023).

Lecturer

Nov 2021 – Oct 2023

Assumption University

Bangkok, Thailand

- Designed and delivered undergraduate computing and engineering courses to 250+ students; maintained 95%+ pass rates across all modules.
- Designed courses and assessments from scratch, aligning content and assessment standards with the Thai Council of Engineers and the Thai Qualifications Framework.
- Supervised two applied deep-learning projects in Thai currency-note detection and traffic-sign recognition, including a team that collected and published a 3,600-image Thai currency dataset on Kaggle; one project member secured a role as Digital Innovation Engineer at TOA upon graduation.
- Served on the AUN-QA committee; defined ELOs, mapped course outcomes to programme outcomes, and trained faculty in Outcome-Based Education practices after completing three AUN-QA professional development workshops.

Teaching Assistant and Tutor

Aug 2016 – Nov 2020

Sirindhorn International Institute of Technology (SIIT), Thammasat University

Thailand

- Conducted laboratory sessions for database programming and microcontroller applications, developing students' practical skills in SQL, PHP, embedded C, and Arduino programming.
- Provided technical troubleshooting support and assisted in course delivery for electrical engineering fundamentals across five consecutive academic years.

Publications

Journal Articles

- 2026** Hussein, S., Salih, R.H., and **Ahmad, I.** "Dynamics and integrability of a modified Sprott C system." *Physica D: Nonlinear Phenomena*, vol. 490, pp. 135160. ISSN 0167-2789. DOI
- 2026** Liu, X., Sang, B., Wang, C., Liu, Y., Yang, C., **Ahmad, I.**, Karimov, T., Rybin, V., Butusov, D., and Wang, N. "Exploring chameleon chaos: A linearly modulated generalized Duffing system with hidden and self-excited attractors." *Int. J. Bifurcation Chaos*, vol. 36, no. 6, pp. 2650074. DOI
- 2026** Mikaeel, S.H., **Ahmad, I.**, and Salih, R.H. "Infinitely many hidden attractors in a simple six-term chaotic 2-torus snap system with a line of equilibria." *Baghdad Science Journal*, vol. 23, no. 4, Article 8. DOI
- 2026** Srisurin, P., **Ahmad, I.**, Ali, N., Khan, R.S., Phuaksuksakul, N., Hussain, Q., and Suparp, S. "A comparative study of predicting travel mode choice of school children using explainable machine learning techniques." *Transportation Research Interdisciplinary Perspectives*, vol. 37, pp. 102035. DOI
- 2025** Wang, C., Sang, B., Liu, X., Wang, C., Liu, Y., Yang, C., **Ahmad, I.**, and Wang, N. "Chameleon chaotic dynamics in a modified Duffing system." *Int. J. Bifurcation Chaos*, vol. 35, no. 10, pp. 2550120. Rank: Q1; IF=2.3. DOI
- 2025** Mikaeel, S.H., Salih, R.H., and **Ahmad, I.** "On the dynamics of a hyperjerk system and its reduction to the generalized Michelson system." *Int. J. Bifurcation Chaos*, vol. 35, no. 13, pp. 2550161. Rank: Q1; IF=2.3. DOI

- 2025** Ahmad, J., **Ahmad, I.**, Miao, Q., and Su, Z. “Forecasting the human cost of disasters under Sustainable Development Goals: A time series analysis using the Facebook Prophet model.” *Technology in Society*, vol. 83, pp. 102992. IF=10.1. DOI
- 2025** Liu, J., Sang, B., Fan, L., Wang, C., Liu, X., Wang, N., and **Ahmad, I.** “Symmetry, Hopf bifurcation, and offset boosting in a novel chameleon system.” *AIMS Mathematics*, vol. 10, no. 3, pp. 4915–4937. Rank: Q1; IF=1.8. DOI
- 2025** Liu, J., Sang, B., Wang, C., Fan, L., Liu, X., **Ahmad, I.**, Karimov, T., Rybin, V., Butusov, D., and Wang, N. “A chameleon system with a cosine function: Bifurcation analysis, multistability, and offset boosting.” *Physica Scripta*, vol. 100, no. 3, pp. 035229. IF=2.9. DOI
- 2023** **Ahmad, I.**, Srisuchinwong, B., and Jamil, M.U. “Coexistence of hidden attractors in the smooth cubic Chua’s circuit with two stable equilibria.” *Int. J. Bifurcation Chaos*, vol. 33, no. 4, pp. 2330010. Rank: Q1; IF=2.3. DOI
- 2021** Liu, M., Sang, B., Wang, N., and **Ahmad, I.** “Chaotic dynamics by some quadratic jerk systems.” *Axioms*, vol. 10, no. 3, pp. 227. Rank: Q2; IF=1.824. DOI
- 2020** **Ahmad, I.** and Srisuchinwong, B. “Simple chaotic jerk flows with families of self-excited and hidden attractors: Free control of amplitude, frequency, and polarity.” *IEEE Access*, vol. 8, pp. 46459–46471. Rank: Q1; IF=3.745. DOI
- 2019** Srisuchinwong, B., Munmuangsaen, B., **Ahmad, I.**, and Suibkitwanchai, K. “On a simple single-transistor-based chaotic snap circuit: A maximized attractor dimension at minimized damping and a stable equilibrium.” *IEEE Access*, vol. 7, pp. 116643–116660. Rank: Q1; IF=4.098. DOI
- 2018** **Ahmad, I.**, Srisuchinwong, B., and San-Um, W. “On the first hyperchaotic hyperjerk system with no equilibria: A simple circuit for hidden attractors.” *IEEE Access*, vol. 6, pp. 35449–35456. Rank: Q1; IF=3.557. DOI

Conference Papers

- 2025** **Ahmad, I.**, Waleed, M., Ullah, A., and Jamil, M.U. “Enhancing credit card fraud detection with hybrid sampling and machine learning models.” *IEEE 6th Int. Conf. on Advancements in Computational Sciences (ICACS)*, Lahore, Pakistan, pp. 1–7. DOI
- 2025** Jamil, M.U., Thanok, S., and **Ahmad, I.** “Design of an IoT-based PLC system for high accuracy color dropping in souvenir machines.” *IEEE 4th Int. Conf. on Electronic Electrical Engineering and Automatic Control (EEEAC)*, Wuhu, China, pp. 154–158. DOI
- 2024** **Ahmad, I.**, Sharif, A.B., Azmat, A., and Jamil, M.U. “A deep learning approach to recognizing new Thai currency using transfer learning for visually impaired individuals.” *IEEE 17th Int. Conf. on Signal Processing and Communication Systems (ICSPCS)*, Gold Coast, Australia. DOI
- 2024** Jamil, M.U., Zahid, M., Chen, Z., **Ahmad, I.**, and Wongthai, P. “Design and implementation of a digital energy monitoring system with enhanced data tracking and analysis features for electrical systems.” *IEEE 2nd Int. Conf. on Power Science and Technology (ICPST)*, Dali, Yunnan, China. DOI
- 2022** Jamil, M.U., **Ahmad, I.**, and Buatoom, U. “Robust adaptive backstepping control design for MIMO electrically driven robot manipulators using RBF neural network with disturbance.” *IEEE 26th Int. Computer Science and Engineering Conference (ICSEC)*, Thailand. DOI
- 2022** **Ahmad, I.**, Jamil, M.U., and Paracha, K.N. “Artificial intelligence-based approach for prediction of power transformer health index.” *IEEE Int. Conf. on Power, Energy and Innovations (ICPEI)*, Pattaya, Thailand. DOI
- 2019** **Ahmad, I.** and Srisuchinwong, B. “On the first single-nonlinearity seven-term memristor-based chaotic snap system: A line equilibrium and coexisting hidden attractors.” *IEEE 24th Int. Conf. on Methods and Models in Automation and Robotics (MMAR)*, Miedzyzdroje, Poland, pp. 292–295. DOI
- 2017** **Ahmad, I.** and Srisuchinwong, B. “A simple two-transistor 4D chaotic oscillator and its synchronization via active control.” *IEEE 26th Int. Symposium on Industrial Electronics (ISIE)*, Edinburgh, UK, pp. 1249–1254. DOI

Teaching Experience

Course outlines, teaching materials, and sample assessments: drirfanahmad.com/teaching

University of Waikato / Hainan University

2025 – Present

- COMPX234: Systems and Networks — COMPX310: Machine Learning
- COMPX326: Computer Vision — COMPX224: Introduction to Databases and Web Development

Maynooth University / MIEC, Fuzhou University **2023 – 2025**

- CS211FZ: Algorithms and Data Structures 2 — EE180FZ: Critical Skills Project
- EE299FZ: Analogue Electronics Project

Assumption University, Bangkok **2021 – 2023**

- CE4114: Data Mining — CE3109: Database Management Systems — CE3104: Software Engineering
- CE2103: Object-Oriented Programming — CE4207: Systems and Network Programming
- CE2709: Computer Architecture — CE3901: Digital Audio/Video Coding Techniques

SIIT, Thammasat University **2016 – 2022**

- CSS326: Database Programming Laboratory — CSS332: Microcontrollers and Applications Laboratory
- ECS204: Electrical Engineering Laboratory (Fall 2016–2020)

Academic Supervision

Bifurcation, Integrability and Hidden Attractors from Chaotic Systems **2022 – Present**
PhD Co-Supervisor, Soran University, Iraq *Thesis writing stage*

Thai Traffic Sign Recognition for Autonomous Cars **2023 – 2024**
B.Sc. Senior Project, Assumption University

A Deep Learning Approach for Thai Currency Note Detection **2022 – 2023**
B.Sc. Senior Project, Assumption University

Honors & Awards

Thammasat University Thesis Publication Grant **2018, 2020**
Awarded three times across doctoral study for independent research publications

Sakura Science Exchange Program in Science **2019**
Tokyo City University, Japan — Fully funded by Japan Science and Technology Agency (JST)

Fully-Funded MSc & PhD Scholarship **2015**
SIIT, Thammasat University, Thailand — Awarded on the basis of academic merit

Gold Medalist — BSc Electrical Engineering **2015**
Highest-ranking graduate of the cohort, Government College University Faisalabad, Pakistan

Review & Editorial Duties

Reviewer, International Journal of Bifurcation and Chaos (IJBC) **2020 – Present**
World Scientific

Reviewer, The European Physical Journal – Plus & Special Topics **2020 – Present**
Springer

Reviewer, IEEE Access **2019 – Present**
IEEE

Certifications

IBM Data Science Professional Certificate **2024**
12-course specialization: Python, SQL, ML, Deep Learning, Generative AI *Coursera / IBM*

Deep Learning A-Z: Neural Networks, AI & ChatGPT Prize **2024**
SuperDataScience Team *Udemy*

Complete Machine Learning, NLP Bootcamp, MLOps & Deployment **2024**
KRISHAI Technologies *Udemy*

Complete Generative AI Course with LangChain and Hugging Face **2024**
KRISHAI Technologies *Udemy*

Statistics for Data Science and Business Analysis **2024**
365 Careers *Udemy*

The Data Science Course: Complete Data Science Bootcamp

365 Careers

2024

Udemy

Machine Learning A-Z: Hands-On Python & R in Data Science

SuperDataScience Team

2021

Udemy

Technical Skills

Programming Languages: Python, MATLAB, Java, R, C++

Machine & Deep Learning: Scikit-learn, XGBoost, LightGBM, CARAT, TensorFlow, Keras, PyTorch, Hugging Face Transformers

Generative AI & LLMs: LangChain, Prompt Engineering, OpenAI API, Retrieval-Augmented Generation (RAG)

Computer Vision: OpenCV, YOLO, Transfer Learning (MobileNet, VGG16, Xception, ResNet)

Data Analysis & Visualization: NumPy, Pandas, SciPy, Statsmodels, Matplotlib, Seaborn, Tableau, Orange, Power BI

Databases & Web: MySQL, SQL Server, PHP, HTML/CSS

Dev Tools: Git, GitHub, Jupyter Notebook, Google Colab, VS Code

Networking & Security: Wireshark, NMAP, Packet Tracer, TCPDUMP, Linux OS, Apache

Instrumentation: LabVIEW, Data Acquisition Systems, Oscilloscopes, PSpice, Multisim, Proteus

Spoken Languages: English (Fluent), Urdu (Native)

References

Dr. Paiboon Sreearunothai

Associate Professor

School of BCET, SIIT, Thammasat University, Thailand

✉ paiboon_sree@siit.tu.ac.th

☎ +66-2-986-9009 (Ext. 3226)

Dr. Abubakar Sharif

Senior Lecturer

University of Waikato, New Zealand

✉ abubakar.sharif@waikato.ac.nz

☎ +86-156-0602-5480

Further references available upon request.